Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

Zhenchao Tang*[1], Fang Wang*†[1], Haohuai He[1], Jiale Zhou[1], Tianxu Lv[1], Jun Zhu[1], Shouzhi Chen[1], Minghao Yang[1], Yu Wang[1], Jiayang Wu[1], Yidong Song[2], Yaokun Li[3], Jiehui Huang[4], Jun Zhou[5], Bing He[1], Jianhua Yao†[1]

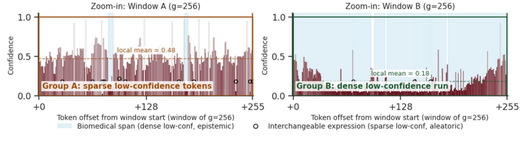
[1]Tencent AI for Life Sciences Lab [2]Nankai University [3]Sun Yat-sen University [4]HKUST [5]Tsinghua University

*Equal contribution †Correspondence: avonwanghit@gmail.com, jianhua.yao@gmail.com

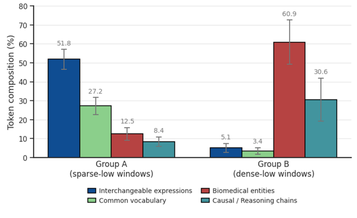
NeurIPS 2026
arxiv.org/abs/2511.21075
shapsider.github.io/BFT

1 Challenge: useful knowledge is difficult

Confidence-based training can suppress unfamiliar biomedical words. An isolated difficult word and a whole difficult passage need different treatment.



2 Motivation: learn difficult passages



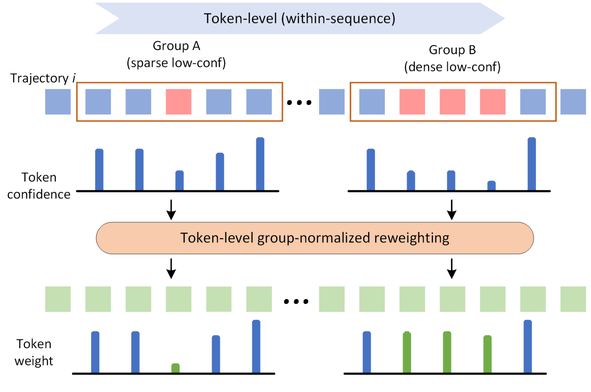
Grouped difficult passages contain more biomedical entities (60.9% vs 12.5%) and reasoning content (30.6% vs 8.4%).

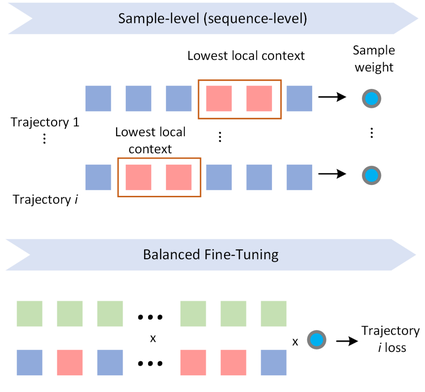
Preserve useful difficulty while retaining broad knowledge.

These associations motivate BFT. They do not identify every uncertain word as useful knowledge.

3 Method: Balanced Fine-Tuning

A simple change to supervised training: use nearby context to decide which words and training examples deserve more learning attention.



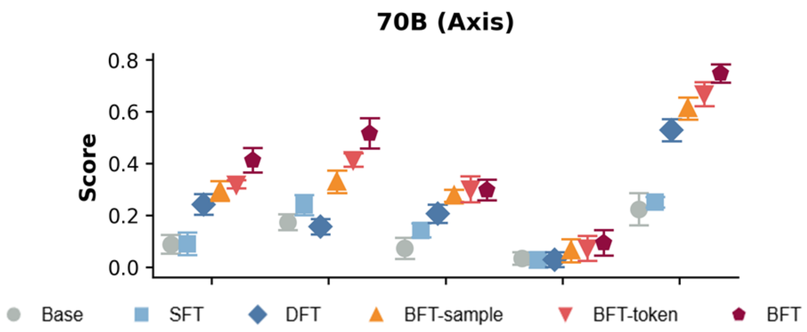


Left: retain words inside difficult passages. Right: emphasize examples containing hard passages, within a limit. BFT uses the model's own confidence, without a reward model or external uncertainty estimator.

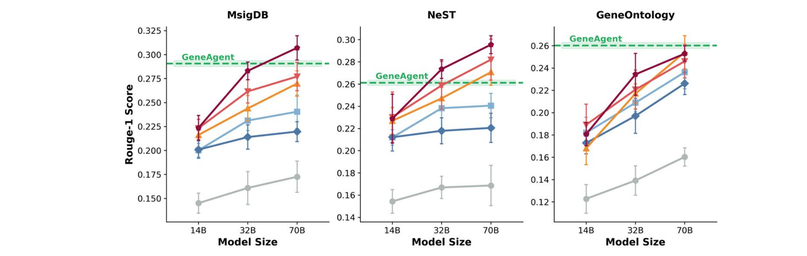
4 Experimental results: biomedical gains with less general forgetting

Stronger biomedical reasoning

HealthBench Hard (70B). Categories, left to right: accuracy, communication quality, completeness, context, instruction following.

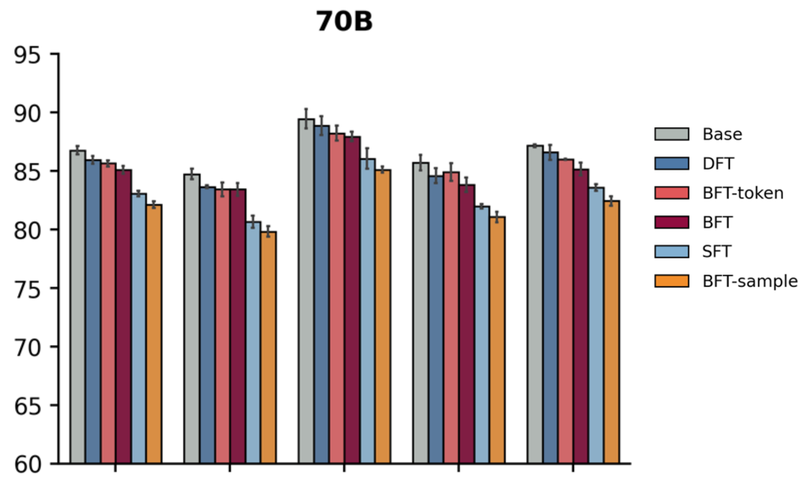


Biological process reasoning



Preserving general capability

MMLU, 70B: BFT preserves more broad knowledge than SFT while adding biomedical capability.



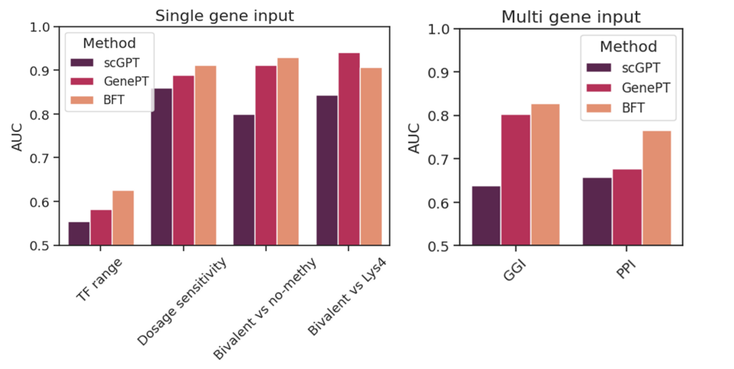
Categories: average, STEM, social sciences, humanities, other.

CMMLU shows the same retention ordering. Some forgetting remains, and DFT forgets least.

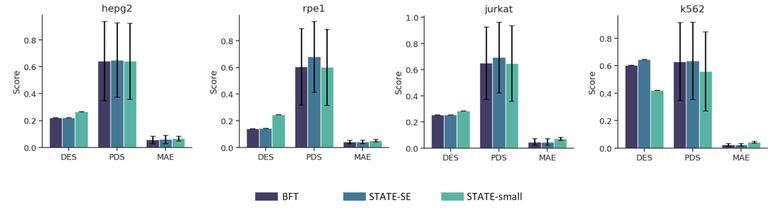
A simple training method for STEM adaptation, demonstrated in medicine and biology.

Useful scientific representations

Gene descriptions become inputs for gene prediction and single-cell response modeling.



Mean gene score: BFT 0.829 vs GenePT 0.802. GenePT remains higher on Lys4. Both use the same text encoder.



Comparable STATE results across four datasets. Profile generation is zero-shot here; the STATE model is still trained.